// Shashank Obla

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# Shashank Obla

**Computer architecture researcher** with comprehensive **cross-stack experience**, from custom digital tapeouts to datacenter-scale stochastic modeling and machine learning. Specialized in the design and optimization of **heterogeneous reconfigurable systems** across **networking, security, and database analytics** workloads.

## EDUCATION

**Carnegie Mellon University, Pittsburgh** — 2019 - 2026 *(Expected)*

Ph.D. Candidate, Electrical and Computer Engineering; GPA: 3.97/4.0 — **Dean's Fellowship**
*PhD Thesis*: Design and Optimization of Input-Dependent Streaming Pipelines on FPGAs
*Advisor*: Prof. James Hoe

**Indian Institute of Technology, Bombay** — 2014 - 2019

B.Tech and M.Tech, Electrical Engineering; GPA: 9.84/10.0 — **Institute Gold Medal**
*Thesis*: Accelerated Circuit Simulation [thesis][paper] | *Advisor*: Prof. Sachin Patkar

## WORK EXPERIENCE

**Intel Corporation** | *Programmable Solutions Group* — Summer 2020

- Modeled **Partial Reconfiguration** (PR) flows for next-generation FPGAs using SystemC to achieve **sub-ms reconfiguration** times utilizing existing hardware resources and to assess the benefits of **new architectural decisions**
- Collaborated with **cross-functional** circuit design teams to analyze the **full-stack** implications of novel PR features

## RESEARCH EXPERIENCE

*preprint available

**RapidQ: Queuing Abstraction and Tuning for Streaming Pipelines on FPGAs*** — 2022 - Present

- Designed the RapidQ abstraction for **performance modeling** of **input-dependent** streaming pipelines on FPGAs
- Devised a modeling framework to extract the model directly from HLS implementations with **>97%** accuracy
- Engineered a **queueing-based** RapidQ simulator in SystemC achieving **>7x speedup** over state-of-the-art
- Developed an **automated tuning** flow powered by RapidQ, **reducing resource utilization by over 40%** across synthetic as well as real-world network security and machine learning workloads

**RapidDetect: Threat Detection via FPGA-Accelerated Log Analysis*** [poster][code] — 2024 - Present

- Developed a **heterogeneous FPGA-CPU** pipeline for streaming log monitoring using Sigma rules at **over 200Gbps**
- Implemented a highly-parametrizable **string matching engine in HLS**, tunable for line rates and log characteristics
- Integrated the FPGA string-matching filter with Hyperscan on CPU achieving **100x reduction** in threat detection latency and **slashing costs by over 10,000x** compared to commercial alternatives using a **single FPGA server**

**Group-by Aggregation Accelerator with NoC-enabled Partitioning** — 2025 - Present

- Architected a streaming group-by aggregation pipeline capable of processing **over 64GBps on a single FPGA**
- Designed and open-sourced a parametrizable Network-on-Chip in RTL operating at **over 500MHz**, which has been used for hash-based tuple partitioning and also in the development of the NoC-based OpenFPGA **by other groups**
- Introduced a fully-associative pre-aggregator **reducing bandwidth by over 2x** for skewed workloads with hot keys

## TECHNICAL SKILLS

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|---|---|
| **Programming** | System Verilog, Vitis/Altera HLS, SystemC, C++, Assembly/Intrinsics, Python \| GitHub Copilot |
| **Design Tools** | Quartus, Vivado, VTR Toolchain, Platform Designer, Cadence - Genus/Innovus/Voltus/Virtuoso |

## OTHER PROJECTS

- Implemented a **hierarchical PR** architecture to dynamically tune modules to changing constraints and opportunities showcasing **>3x improvement** in performance for a Breadth-First Search accelerator over static provisioning [pdf]
- Designed and taped out a **200MHz FPGA fabric** with a VTR-based software toolchain on **TSMC 28nm** process
- Accelerated a stencil-based scientific computing workload achieving **>140x speedup** using **SIMD intrinsics** [pdf]

## LEADERSHIP

- Mentored 20+ PhD students in **applying queuing theory** to computer systems research with Prof. Harchol-Balter
- Chaired the Student Council for Faculty Hiring, **managing** a group of **25 students** in conducting faculty **interviews**